

CS-302: Artificial Intelligence

LECTURE NO 3:

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Organization of Lecture

1. Classification of Environment in Artificial Intelligence
 1. Accessible and Inaccessible Environment
 2. Deterministic or Non-deterministic Environment
 3. Static or Dynamic Environment
 4. Discrete or Continuous Environment
 5. Observable or Partially Observable

1. Classification of Environment in AI (1/3)

Classification of Environment in AI:

→ Environment is part of the universe that surrounds Intelligent system.
→ Environment + Agent = Universe.

① Accessible and Inaccessible

- Agent can obtain complete and accurate information about state's environment. (Accessible) ←
- Whole part of the environment may not be in reach. (Inaccessible)
- Example of Accessible environment: Room with temperature as state.
- Example of Inaccessible environment: Any event on earth.

② Deterministic or Non-Deterministic

- Agent's current state and selected action can completely determine environment next state (Deterministic)
- Nature of environment can't be decided by agent alone (Non-deterministic/stochastic).
- Example of deterministic environment: Agent doesn't have to worry about uncertainty.
- Example of non-deterministic environment: Random in nature.

1. Classification of Environment in AI (2/3)

③ Static or Dynamic

- ↳ Environment doesn't change its state with passage of time (Static).
- ↳ State of environment changes with respect to time (Dynamic).
- ↳ Example of static environment: crossword puzzles.
- ↳ " " dynamic " : Car driving.

④ Discrete or Continuous

- ↳ Finite no. of percepts and actions (Discrete), for example: Chess Game
- ↳ Indiscrete environment observations are not continuous.
- ↳ Agent can perceive and make observations from the environment continuously without any lag (Continuous), for example: Self-driving car

1. Classification of Environment in AI (3/3)

⑤ Observable or Partially observable

↳ Agent sensor can access complete state of environment at each point of time. (observable), It is fully or completely observable

↳ In partially observable, some part of environment is not in reach of agent.

References: Textbooks

1. *“Artificial Intelligence”, by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
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3. *“Computational Intelligence: A logical Approach”, by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *“Artificial Intelligence: Structures and Strategies for Complex Problem Solving”, by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *“AI: A new Synthesis”, by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *“Artificial Intelligence: Theory and Practice” by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*